Normal Forms Summary

DB normal forms



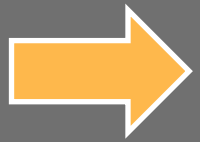
- Domain/key
- 6th
- 5th = Project-join
- 4th
- Boyce-Codd = 3.5^{th}
- 3rd
- 2nd
- 1st

Based on functional dependencies, i.e., what attributes depend upon.

Goal: Prevent DB anomalies.

1st normal form

instructor	days_teaches
John	{M,T,W,Th,F}
John	{M,T,W,Th,F}
Scott	{T,Th}
…	…



instructor	day_teaches
John	M
John	T
John	W
John	Th
John	F
Scott	T
Scott	Th
…	…

Table represents a mathematical relation:

- Each tuple unique.
- Single value per tuple & attribute.

4NF – multi-valued dependencies (generalizes FDs)

5NF & 6NF – join dependencies

DKNF – only domain & key constraints

GLIMPSE BEYOND BCNF/3NF/FDS

Multi-value dependencies

FD: $A \rightarrow B$ The value in A determines a value for B.

MVD: $A \twoheadrightarrow B$ The value in A determines a set of values for B.

restaurant	menu_item	delivery_area
Papa John's	Pizza	Rice Village
Papa John's	Pizza	Rice University
Papa John's	Pizza	Southampton
Domino's	Pizza	Rice Village
Domino's	Pizza	Rice University
Domino's	Pasta	Rice Village
Domino's	Pasta	Rice University

{restaurant} ↠ {menu_item}
{restaurant} ↠ {delivery_area}

Normal form summary

- NF constrains table design to limit redundancy.
- Have decomposition algorithm to normalize.
- Different NFs offer different trade-offs.
  - BCNF most common.
  - Sometimes 3NF.
  - Occasionally 4NF.